

Case Study: Sentiment and Thematic Analysis of Shipping Companies' Customer Reviews

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Purpose

This case study emerges you into the shipping industry and asks you to act as a consultant by analyzing customer reviews and making suggestions, specifically for USPS, UPS, FedEx, and DHL. You will work on developing a machine learning model to conduct a sentiment analysis and thematic analysis on shipping companies' performance. You will work with real customer reviews and analyze them using Natural Language Processing (NLP) and Latent Dirichlet Allocation (LDA). This is an opportunity to build practical experience with using data and various models to compare businesses and make decisions rooted in data. By the end of the project, you will have gained insights into how machine learning supports businesses and facilitates improvement based on customer reviews. In the real world, consulting begins with data collection, so this case study asks you to be involved in not only data collection, but analysis, and recommendations. To ensure the comparison and recommendations you give are the best, you will need to navigate dataset limitations and technical challenges.

The Task

Your task is to step into the role of a data scientist consultant tasked with deciding who is the superior shipping company based on customer reviews sentiments and what areas each business could use improvement in based on the reviews. Using a provided dataset of 120 reviews, you will:

1. **Collect More Data:** Collect an additional 30 customers reviews from the most recent Trust Pilot website for each company so have a larger range of data across time.
2. **Build a Machine Learning Model:** Leverage Natural Language Processing for sentiment analysis and Latent Dirichlet Allocation for the customer review themes. Split the data into training, validation, and test sets.
3. **Evaluate and Compare the Models' Results:** Using the sentiment scores to compare the results from each company. Compare the frequency of themes for each company.
4. **Draw Conclusions:** Based on the evaluation, determine who is the best and worst performing shipping company and why (based on the themes represented most).
5. **Make Recommendations:** Based on the results, make tailored recommendations for improvement for each company.
6. **Reflect:** Reflect on the challenges you encounter, such as limited dataset, technical difficulties, and what next steps would be to improve the model or any part of the process.

GitHub Link: <https://github.com/simoneminorr/Case-Study-Three/tree/main>

You will document your work in a structure report detailing the strengths and weaknesses of each company along with recommendations for improvements for each company. Additionally, you will attach the datasets, scripts, and visualizations as well as corresponding documentation about the steps taken throughout the process and the limitations in your work. You will upload all of the necessary files into your own GitHub repository.